WESTERN PACIFIC UNIVERSITY

CMN224/CMP224 Software Engineering

Task 2: Requiremnts & Estimation Report

Project: Kina Ridge Primary School Fee – Tracking System

Scenario: **A**
Name: **Justin Albert**
Student ID: **00110**
Lecturer: **Mr. Solomon. Hombiku**
Due date: **Week 8**

Part 1: Concise Software Requirements (SRS)

1.1 Purpose and Scope

1.1.1 Purpose

Kina Ridge Primary School is a Port Moresby private institution that has approximately 450 students. The school is presently keeping track of fees with a notebook and Excel. Parents have the option to pay with cash, bank deposit, or mobile transfer, and certain payments may be made in installments.
The existing method restricts the ability to locate precise payment records, recognize students with unpaid fees, and issue reliable receipts.
The aim of the suggested Kina Ridge Primary School Fee-Tracking System is to establish an easy and dependable method for documenting fees and payments, monitoring balances, generating receipts, and creating fundamental reports.

1.1.2 In Scope

The first version of the system will include:

- Student and fee records.
- Payment recording.
- Cash, bank deposit and mobile-transfer payment methods.
- Instalments payments
- Automatic outstanding-balance calculation.
- Payment history and student search.
- Receipt generation and printing.
- Arrears identification.
- Basic management reports.
- User login and assess control
- Basic audit trial and data backup.

1.1.3 Out of Scope

The following features are outside the scope of the first version:

- Student attendance or examination management.
- Payroll or human-resource management.
- Online teaching.
- School timetable management.
- Full accounting software.
- Direct bank or mobile transfer integration.
- Online payments.
- Parent mobile transfers or SMS services.
- Stock management.

By keeping all these features outside the first release helps control project scope, time and cost.

1.2 Business context and Stakeholders

Kina Ridge Primary School enrolls around 450 students. Fee details are presently separated between a notebook and an Excel spreadsheet. Payments can be made through various methods and can be done in installments.

The bursar is the primary user, so the system needs to be straightforward and effective for everyday operations. The principal and board primarily need reports. Parents and guardians require precise records, as inaccurate payment details can result in conflicts.

Stakeholder	Role	Main Need
Bursar	Main system user	Needs to quickly record payments, check balances and issue receipts.
Office assistants	Supporting users	Need to search student records and help record or confirm payments.
Head teacher	Management user	Needs accurate reports about payments and fee arrears.
School board	Management/oversight	Needs term summaries/reports to understand the school's fee collection position.
Parents/guardians	Payers	Need accurate records and trustworthy receipts for payments made.
Students	Beneficiaries of school administration	Need their fee records to be correctly maintained.
School management	Business owner	Needs a reliable system that reduces errors and missing records.
System administrator	Technical/support role	Needs to manage user access and backups.

1.3 Functional Requiremtns

Functional requirements describe what the system shall do.

The requirements below use MoSCoW priorities:

- **Must** - important for the first release.
- **Should** - important but the system can operate without it initially.
- **Could** - useful if time and budget allow.
- **Won't** - purposily excluded from the first release.

ID	Functional Reequirement	Priority
FR-01	The system will enable an authorized user to generate a student fee record.	Must
FR-02	The system will keep the student's name, student ID, and associated class details.	Must
FR-03	The system must enable an authorized user to record the fees that a student owes.	Must
FR-04	The system shall allow authorised users to record cash payments.	Must
FR-05	The system must enable authorised users to record bank deposit payments.	Must
FR-06	The system shall allow authorised users to record mobile-transfer payments.	Must
FR-07	The system shall allow a payment to be recorded as an installment.	Must
FR-08	The system shall automatically calculate the remaining balance after a payment is recorded.	Must
FR-09	The system shall display the payment history for an individual student.	Must
FR-10	The system shall generate a unique receipt/reference number for each recorded payment.	Must
FR-11	The system shall allow authorised users to print a payment receipt.	Must
FR-12	The system shall allow users to search for a student by student ID or name.	Must
FR-13	The system shall identify students with outstanding fee balances.	Must
FR-14	The system shall allow authorised users to correct an incorrectly entered transaction while keeping a record of the change.	Should
FR-15	The system shall produce a report showing total fees collected for a selected period.	Should
FR-16	The system shall produce a report showing students with outstanding balances.	Should
FR-17	The system shall produce a term summary report for school management and the board.	Should
FR-18	The system shall allow authorised users to view daily payment totals.	Should
FR-19	The system could provide simple charts showing fee collection and arrears.	Could
FR-20	The system could allow export of selected reports to a common file format such as PDF or spreadsheet format.	Could
FR-21	The system shall not provide payroll, attendance or student examination management in the first release.	Won’t

Key Functional Priorities

The most important functions are payment recording, balance calculation, receipts and arrears information. These functions are prioritised as **Must** because they directly address the main problems described by the client. More advanced functions such as charts are lower priority because they are useful but are not necessary for the school to manage fees.

1.4 Non-functional Requirements

Non-functional requirements describe how well the system must work and the conditions under which it must operate.

The requirements below specifically consider the conditions at Kina Ridge Primary School.

ID	Non-Functional Requirement	Measure/Reason
NFR-01	The system shall save a payment before showing it as successfully recorded.	No unsaved payment shall appear as successful.
NFR-02	The system shall display a student record within 3 seconds under normal conditions.	Staff should not have to wait a long time for records.
NFR-03	A trained bursar shall be able to record a normal payment within 2 minutes, excluding printer delays.	Supports fast daily payment work.
NFR-04	The system shall require user login before financial records can be accessed.	Unauthorised users cannot access records.
NFR-05	Only authorised users shall change payment records.	Access is controlled by user role.
NFR-06	The system shall record the user and date for important changes to financial records.	Changes can be traced.
NFR-07	Saved transactions shall remain available after a power interruption and restart.	No loss of successfully saved transactions.
NFR-08	The system shall operate on the school's existing office laptop.	No specialised computer required.
NFR-09	The system shall support printing of receipts using the school's available printer.	Printed receipts are required for trustworthy payment evidence.
NFR-10	Essential record entry should remain possible during temporary internet loss where the chosen technical solution supports it.	The system should not depend unnecessarily on continuous internet access.
NFR-11	Financial records shall be backed up at least once per working day.	Reduces the risk of losing financial records.
NFR-12	The system shall use simple labels and forms suitable for trained office staff.	The bursar and office assistants are the main users.
NFR-13	Money values shall be displayed in Papua New Guinea kina.	Currency shown as K/PGK.

NFR-14	The system shall prevent a user from recording a payment with an invalid or zero amount.	Reduces data-entry errors.

PNG Delivery Conditions

The school experience weekly power outages, thus stored records must endure these power disruptions. There is just one primary office laptop, so the system must be compatible with the current hardware. A printer is ready and needs to enable receipt printing.
WiFi typically functions, but ongoing internet availability shouldn't be considered assured. The system should accurately represent the actual payment methods of the school by accommodating cash, bank transfers, mobile payments, and installment plans.

1.5 Constraints and Business Rules

Constraints

The following constraints are based on the client scenario:

1. The school has approximately 450 students.
2. The school currently has one office laptop for the main fee-recording work.
3. A printer is available.
4. Power cuts occur weekly.
5. WiFi is usually available but should not be assumed to be available at all times.
6. The school currently uses an exercise book and Excel.
7. Payments may be made using cash, bank deposits or mobile transfer.
8. Parents may pay fees in instalments.
9. The first release has a limited project scope.
10. The project should use existing school equipment where practical.

Business Rules

BR-01: Every payment shall belong to a valid student record.
BR-02: Every payment shall have a date.
BR-03: Every payment shall have a payment method.
BR-04: The payment methods supported in the first release shall be cash, bank deposit and mobile transfer.
BR-05: A student may have multiple payments because fees can be paid in instalments.
BR-06: The student's outstanding balance shall be calculated from the fee amount and recorded payments.
BR-07: The system shall not allow an invalid payment amount to be recorded.
BR-08: Every successful payment shall receive a unique receipt/reference number.
BR-09: A payment record shall not be silently deleted after it has been used for financial reporting.
BR-10: Corrections to financial records shall be restricted to authorised users.

BR-11: Important changes to financial records shall be traceable to the user who made the change.
BR-12: Only authorised staff shall access student financial information.
BR-13: A payment shall not reduce the student's balance below zero unless the school later approves an overpayment policy.
BR-14: The first release shall not include full accounting, payroll or student management functions.

1.6 Glossary

Term	Definition
Arrears	Fees that remain unpaid after they are due.
Balance	The amount a student still owes.
Bursar	Staff member responsible for school financial records.
Instalment	A partial payment of a larger fee amount.
Mobile money	Payment made through a mobile-money service.
Receipt	Evidence that a payment has been recorded.
Payment reference	An identifier used to identify a payment.
Functional requirement	Something the system must do.
Non-functional requirement	A condition describing how the system must operate.
Stakeholder	A person or group affected by the system.
MoSCoW	Method for prioritising requirements as Must, Should, Could or Won't.
Audit trail	Record of important changes showing who made them and when.
Backup	A copy of data used to recover lost information.
Scope	The work and features included in the project.

Part 2 - Estimate & work schedule

2.1 Task Breakdown

The project is divided into smaller tasks so that each part can be estimated. The breakdown also helps identify dependencies between activities.

ID	Task	Main Work
T1	Requirements	Confirm needs and finalise SRS.
T2	Planning	Plan system structure and technical approach.
T3	Database design	Design student, fee, payment and user data.
T4	Interface design	Design simple user screens.
T5	User access	Develop login and permissions
T6	Student/fee module	Develop student and fee records.

T7	Payment module	Develop cash, bank, mobile-money and instalment recording.
T8	Balance/receipt functions	Develop balance calculation and receipts.
T9	Search/reports	Develop search, history, arrears and management reports.
T10	Audit/backup	Develop change tracking and backup process.
T11	Testing	Test functions, calculations, security and recovery.
T12	User acceptance	School staff test the system.
T13	Deployment/training	Install system, train users and hand over.

2.2 Time Estimates, Assumptions and Reasoning

Estimation Method

A **three-point estimation approach** is used for this initial estimate. For each major task, three values are considered:

- **Optimistic (O):** The task goes smoothly.
- **Most likely (M):** Normal project conditions occur.
- **Pessimistic (P):** Problems, rework or delays occur.

The expected estimate is calculated using:

Expected Time = (O + 4M + P) / 6

This method is useful because software work is uncertain. For example, requirements may need clarification, testing may find defects, or the client may need additional time to review a feature.

Estimate Table

Task	O	M	P	Expected Hours
T1 Requirements	8	10	14	10.3
T2 Planning	4	6	9	6.2
T3 Database design	6	8	12	8.3
T4 Interface design	5	8	12	8.2
T5 User access	4	6	9	6.2
T6 Student/fee module	8	12	18	12.3
T7 Payment module	12	18	27	18.5
T8 Balance/receipts	8	11	16	11.3
T9 Search/reports	8	12	11	7.3
T10 Audit/backup	5	7	11	7.3
T11 Testing	12	18	27	18.5
T12 User acceptance	6	8	12	8.3
T13 Deployment/training	6	8	12	8.3

Total **126.0 hours**

The estimated effort is approxmatley **126 hours.**

The estimate is not presented as an exact prediction. It is a planning estimate based on the defined scope and assumptions.

Estimation Assumptions

The estimate assumes:

1. One main developer.
2. A small system for approximately 450 students.
3. The client provides information and feedback on time.
4. The existing laptop and printer can be used.
5. No direct bank or mobile-money integration is required.
6. No major new requirements are added after approval.
7. Staff are available for testing and training.
8. No major hardware purchase is required.
9. Power interruptions are considered in system testing.
10. The project uses the scope defined in this report.

Reasoning

The **payment module** receives the largest development estimate because it must handle cash, bank deposits, mobile transfer and instalments.

Testing also receives a large estimate because financial information must be accurate. Errors in balances or receipts could cause disputes.

Requirements and design are completed before major development so that the system is based on the school's actual process.

2.3 Work Schedule

The estimated **126 hours** can be organised into an approximately **8-week schedule**, assuming the project is carried out part-time and some tasks overlap.

Week	Work	Client Sees
1	Requirements and SRS	Confirmed requirements and priorities.
2	Planning, database and interface design	Basic system design and screens.
3	User access and student/fee module	Login and student records.
4	Payment module	Payment entry and instalments.
5	Balance, receipts and search	Balances, receipts and payment history.
6	Reports, audit and backup	Arrears and management reports.
7	Testing and corrections	Test version for staff

8	User acceptance, deployment and training	Working system and trained users.

Delivery Order

- The work is planned in this order because some activities depend on earlier activities.
- Requirements must be confirmed before the system is fully designed.
- The database and interface design should be completed before the main development work.
- Student and fee information should exist before payment transactions can be recorded.
- Payment recording must be completed before reliable arrears and reports can be produced.
- Testing must occur before deployment.
- User acceptance testing should happen before the final handover so that the bursar can confirm that the system supports real school processes.

Main Milestone

Milestone 1: Requirments Approved
The school confirms that the SRS correctly describes the required system.

Milestone 2: Basic Working System
User can log in and manage basic student and fee information.

Milestone 3: Core Fee Tracking Complete
Users can record payments, including instalments, and see outstanding balances.

Milestone 4: Reporting and Receipt Functions Complete
Receipts, arrears and management reports are available.

Milestone 5: Acceptance and Handover
The bursar and office assistants test the system, staff receive training and the system is prepared for normal use.

2.4 Dependency-Triangle Analysis

The project has three important factors:

Scope — Time — Cost

These factors are connected. If one changes, at least one of the other factors normally has to change.

Current Recommended Position

The recommended first release should focus on the **Must-have requirements**:

- Student fee records.
- Payment recording.
- Cash, bank and mobile-money payments.
- Instalment payments.
- Balance calculation.
- Receipts.
- Student search.
- Payment history.
- Arrears identification.
- Security.
- Backup.
- Basic reporting.

This scope directly addresses the school's main problems.

Advanced features such as charts and automatic integrations should not delay the core system.**Option 1 — Keep the Scope and Keep the Quality**

Under this option, the school receives the planned first release.

Estimated effort: approximately 126 hours.

Expected schedule: approximately 8 weeks.

Cost: moderate and based mainly on developer effort.

The advantage is that the school receives the important features without deliberately reducing the agreed scope.

This is the recommended option.

Option 2 — Finish Faster

If the school says it must have the system in a much shorter period, the project could use additional development resources.

For example, two developers could work on different parts of the system.

However, adding developers does not automatically reduce the project time by half because some activities still depend on each other. More developers can also create additional coordination and testing work.

This option is likely to **increase cost**.

The school could also reduce the first-release scope to make the deadline easier to meet.

For example, the first release could delay:

- Charts.
- Advanced report export.
- Some non-essential reporting features.

The core payment and receipt functions would remain.

Option 3 — Reduce Cost

If the school has a strict budget, the project could reduce the number of features in the first release.

The first release could contain only:

- Student records.
- Fee records.
- Payment recording.
- Instalments.
- Balance calculation.
- Receipts.
- Arrears.
- Basic reports.
- Security and backup.

Charts and advanced report functions could be added later.

The disadvantage is that the school would receive fewer features initially.

Another possibility would be to keep the same scope but allow a longer development period.

Dependency-Triangle Comparison

Option	Scope	Time	Cost
1. Balanced	Core + important features	About 8 weeks	Moderate
2. Fast delivery	Reduced scope or more developers	Shorter	Higher if extra resources are used
3. Lower cost	Smaller first release or longer schedule	Longer or reduced features	Lower

Recommendation

Option 1 is recommended.

The school specifically wants **fee arrears to be visible and receipts to be trustworthy**. Therefore, the first release should focus on accurate payment records, instalments, balances, receipts and arrears.

Advanced features should not delay these core functions. They can be added in a later release if the school has additional time and budget.

This approach provides a reasonable balance between **scope, time and cost** and reduces the risk of delivering an overly complicated system.

Conclusion

The Kina Ridge Primary School Fee-Tracking System aims to substitute the existing combination of an notebook and Excel Spread sheet with a more dependable and structured method for recording fees.

The system will prioritize the school's key requirements: documenting various payment methods, accommodating installment plans, computing balances, generating reliable receipts, and detecting outstanding payments. The requirements also take into account the PNG delivery conditions of the school, such as weekly power outages, restricted hardware, availability of printers, WiFi constraints, and the various methods parents use to pay fees.

The project has been segmented into 13 key tasks and assessed through three-point estimation. The anticipated effort is around 126 hours, with a delivery timeframe of about 8 weeks. The suggested strategy is to implement the main fee-tracking functionalities initially and delay non-critical features. This provides the school with an effective system while maintaining control over the project's scope, schedule, and budget.

AI Use Declaration: I use AI as a support tool, mainly to help organise my ideas, improve the wording and structure, and check whether I had covered the requirements and estimation sections myself. I provided the project scenario and reviewed the suggestions myself. I made the final decisions about the requirements, estimates, assumptions and scope. I used AI to assist my work, not to replace my understanding of the assignment.